Project Proposal: Text-to-Image Generation with CLIP and Stable Diffusion

IE7615 Deep Learning for AI | Group 1, Section 03 | Generative project | Date: November 5, 2026

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1. Objective

Build a text-to-image pipeline that turns natural-language prompts into 512×512 images, using a curated COCO image–caption subset to (a) establish a reproducible zero-shot baseline with Stable Diffusion v1.5 and (b) measure whether lightweight LoRA fine-tuning improves prompt adherence on COCO-style scenes.

2. Dataset

- **Source:** MS-COCO 2017 captions (official annotations, CC BY 4.0), 5 human captions per image across everyday objects and scenes.
- **Curated subset:** 1,998 image–caption pairs (2,000 sampled, 2 duplicate captions removed), seeded random sample, split 1,598 train / 199 val / 201 test by image so no image crosses splits.
- **Cleaning:** ASCII normalisation, whitespace and punctuation fixes, drop captions under 4 or over 40 words (CLIP caps at 77 tokens), remove duplicate captions, require shorter image side ≥ 256 px; square centre-crop to 512×512.
- **Why COCO over Flickr30k:** broader object and scene variety, direct download from official URLs, standard benchmark for CLIP-based evaluation. Flickr30k is people-centric and has stricter access terms.

3. Model and approach

- **Architecture:** Stable Diffusion v1.5 latent diffusion. A frozen CLIP ViT-L/14 text encoder produces 77×768 token embeddings; a UNet denoises in the VAE latent space with cross-attention to the text.
- **Baseline (Phase 1):** zero-shot SD v1.5, DPM-Solver++ with 30 steps, guidance scale 7.5, five fixed prompt–seed pairs so every later model is compared on identical inputs.
- **Adaptation (Phase 2):** LoRA on UNet attention layers (rank 8–16) trained on the 1,598 train pairs in fp16. Ablations: single vs. all five captions per image, and LoRA rank.
- **Stack and compute:** PyTorch, Hugging Face Diffusers, Transformers (CLIP); fits a 16 GB GPU (Colab T4); GitHub for version control.

4. Evaluation

- CLIPScore (prompt adherence) on the 5 test prompts and on the 201 held-out test captions; text→image retrieval recall on val as a data-quality check.
- FID/KID of generated vs. real test images (noted as noisy at n≈200), plus a small blind human preference comparison of baseline vs. LoRA outputs.
- **Deliverables:** this proposal, 5 sample prompt→image test runs, and an initialised GitHub repo with data pipeline, baseline notebook and tests.

5. Team roles

Role	Owner	Responsibilities
Data lead	Aditi Kulkarni	Dataset curation, cleaning and preprocessing pipeline; split and dataset statistics.
Modeling lead	Aditya Sai Kumar Kusetti	Environment, SD v1.5 baseline, LoRA fine-tuning runs, hyperparameter ablations.
Evaluation lead	Sakshi Pawar	CLIPScore, retrieval and FID/KID metrics, fixed-seed prompt panel, human preference check.
Integration & docs lead	Hemanth Mareedu	GitHub repo, tests, README, results tracking, final report.